# Task 3: Improvement, Ablation, Cross-Validation and Explainability of a Laterality-Aware Spatio-Temporal Graph Neural Network for 64-Channel EEG Motor Movement Classification

---


Asfar Hossain Sitab
ID-2026-2-74-008

Parmita Hossain Simia
ID-2026-2-74-009

Supervisor: Raihan Ul Islam
Associate Professor, Dept. of CSE
East West University (EWU)



8 August 2026


---

**Abstract**

Task 3 investigates whether the weak first-generation Hybrid ST-GNN from Task 2 can be improved through controlled ablation, a laterality-aware redesign, subject-independent cross-validation, statistical testing, and local explainability. The executed notebook retained 4,548 accepted left/right-fist motor-execution trials from 108 subjects and used the 91-subject Task-2 development population for the primary five-fold grouped evaluation. Twenty-three single-factor ablations varied GNN depth, hidden width, dropout, batch normalization, learning rate and schedule, optimizer, graph construction, edge weighting, GNN family, class-imbalance handling, and early stopping. The highest fixed-split validation macro-F1 was obtained by `A09_lr_0015` (0.6296), which kept the two-layer GCN and weighted hybrid graph but increased the learning rate to $1.5 \times 10^{-3}$. Under five-fold subject-disjoint CV, the final laterality-aware ST-GNN reached accuracy $0.5908 \pm 0.0204$ and macro-F1 $0.5877 \pm 0.0196$. This was a clear improvement over the first Hybrid ST-GNN (macro-F1 $0.4937 \pm 0.0161$), but it remained below the 1D-CNN baseline (macro-F1 $0.6920 \pm 0.0221$). At the subject level, the final GNN minus 1D-CNN mean accuracy difference was $-0.1048$ with a bootstrap 95% confidence interval of $[-0.1287, -0.0812]$ and a paired sign-flip randomization-test $p = 0.000020$. SHAP and a LIME-style local surrogate were then applied to one correct and one incorrect prediction while the raw EEG was held fixed, revealing distributed rather than narrowly localized channel reliance. The final conclusion is therefore improvement over the first GNN, not superiority over the best baseline.

**Keywords:** EEG; graph neural network; ablation study; subject-independent cross-validation; statistical significance; SHAP; LIME; motor execution; leakage prevention.

# Contents

# 1. Task 3 Objective

Task 2 established a leakage-conscious left-versus-right fist motor-execution benchmark on the PhysioNet EEG Motor Movement/Imagery Dataset and showed that the first Hybrid ST-GNN generalized poorly to unseen subjects. Task 3 therefore focuses on diagnosis and controlled improvement rather than simply increasing model complexity. The required objectives were: (i) perform ablations over architecture, optimization, graph design, imbalance handling, and stopping policy; (ii) select one final GNN without tuning on a final evaluation fold; (iii) compare the final GNN against the strongest baseline and the first GNN using five-fold subject-level CV; (iv) conduct a paired significance test at the subject level; and (v) explain one correct and one wrong prediction with SHAP and LIME.

The notebook completed all requested experiment families. Importantly, the scientific claim tested here is not "GNNs are better than CNNs." The defensible question is whether the redesigned graph model improves on the first graph model, and whether any remaining difference relative to the strongest baseline is statistically and practically meaningful.

# 2. Data, Evaluation Population, and Pillar-A Audit

## 2.1. Dataset retained from Task 2

The input tensors loaded by the Task 3 notebook contain 4,548 accepted trials from 108 subjects. Each trial contains 64 EEG channels and 400 post-cue samples; the handcrafted node-feature tensor contains ten features per channel. The class distribution remains almost perfectly balanced: 2,276 left-fist and 2,272 right-fist trials.

Task 3 does not reuse the Task-2 test split as the primary evidence pool because that test set has already been inspected in the earlier report. Instead, the Task-2 training and validation partitions are combined to form a development population of 3,840 trials from 91 subjects. This population is used for the five-fold grouped comparison.

## 2.2. Two-stage evaluation logic

The executed workflow has two stages:

1. **Stage A—controlled ablation.** Each single-factor ablation is trained on the Task-2 training split and ranked using Task-2 validation macro-F1. The Task-2 test labels are excluded from ablation selection.

2. **Stage B—five-fold grouped evaluation.** The selected final configuration, the 1D-CNN baseline, and the first Hybrid ST-GNN are evaluated using five outer GroupKFold partitions over the 91-subject development population. Within every outer-training fold, a subject-disjoint inner validation split is created for early stopping. Preprocessing statistics and graph construction are fitted using inner-training data only.

The resulting outer-fold audit is shown in Table 1. Every outer train/test subject overlap is zero.

Table 1: Subject-disjoint five-fold outer-CV audit.

| Fold | Train trials | Test trials | Train subjects | Test subjects | Overlap |
|---|---|---|---|---|---|
| 1 | 3073 | 767 | 73 | 18 | 0 |
| 2 | 3075 | 765 | 73 | 18 | 0 |
| 3 | 3060 | 780 | 72 | 19 | 0 |
| 4 | 3076 | 764 | 73 | 18 | 0 |
| 5 | 3076 | 764 | 73 | 18 | 0 |

### 2.3. Pillar-A fairness note

Within every outer fold, there is no subject overlap and the outer test subjects do not contribute to fold-specific normalization, functional graph estimation, class weighting, early stopping, or checkpoint selection. This satisfies the central subject-separation requirement of Pillar A.

A stricter methodological caveat must nevertheless be stated. The final hyperparameter configuration was chosen earlier on the fixed Task-2 validation subjects, and those subjects are subsequently part of the 91-subject CV population. Therefore, the reported five-fold CV is *subject-disjoint at model fitting and testing*, but it is not a fully nested CV in which hyperparameter selection is repeated entirely inside every outer fold. The present results are suitable as a controlled comparison of fixed configurations; a publication-grade final estimate should use nested grouped CV or keep the original tuning subjects outside all outer evaluation folds.

## 3. Improved Laterality-Aware ST-GNN

### 3.1. Design changes

The Task 3 model explicitly introduces electrode identity and left/right structure instead of relying only on repeated graph propagation. A shared channel-wise temporal encoder applies two 1-D convolutional stages to each electrode. In parallel, the ten handcrafted node features are projected to the same hidden dimension. The model then concatenates four sources of node information:

$$z_i = \mathrm{Fuse}\left[z_i^{(\text{temporal})}, z_i^{(\text{feature})}, e_i^{(\text{identity})}, s_i^{(\text{hemisphere,motor})}\right]. \tag{1}$$

The fused node representation is processed by configurable GCN, GraphSAGE, or GAT layers. The final readout is deliberately laterality-aware: attention pooling, global maximum pooling, left-hemisphere mean, right-hemisphere mean, the explicit left-minus-right difference, and a midline mean are concatenated before classification. This preserves a direct path for hemispheric asymmetry, which is physiologically relevant for left/right motor tasks.

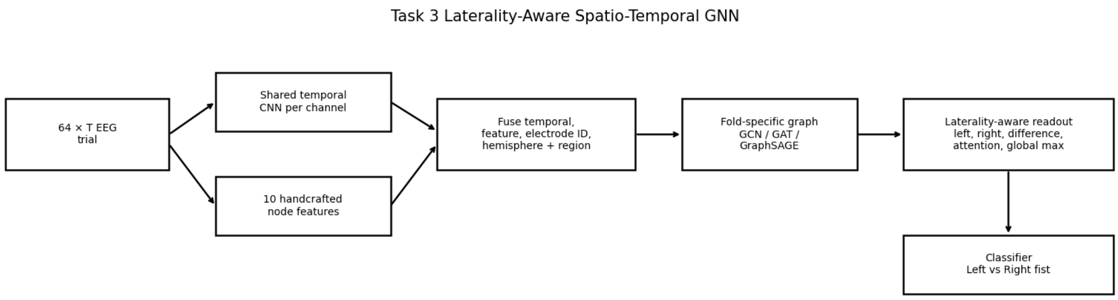



Figure 1: Task 3 laterality-aware ST-GNN architecture implemented in the supplied notebook.

### 3.2. Selected configuration

The reference ablation used a two-layer GCN with hidden size 64, dropout 0.30, batch normalization, AdamW, cosine decay, a weighted hybrid anatomical-functional graph, class weighting, and early stopping. The best single-factor experiment was `A09_lr_0015`; only the learning rate changed to $1.5 \times 10^{-3}$. The selected model therefore retained the following settings:

Table 2: Final Task 3 configuration selected by validation macro-F1.

| **Component** | **Selected setting** |
|---|---|
| GNN family / depth | GCN, 2 graph layers |
| Hidden size | 64 |
| Dropout / batch normalization | 0.30 / enabled |
| Optimizer | AdamW |
| Learning rate / schedule | $1.5 \times 10^{-3}$ / cosine |
| Graph | weighted hybrid; 0.40 anatomical + 0.60 functional |
| Functional neighbours | 8 strongest training-only absolute correlations per node |
| Imbalance handling | class weights |
| Early stopping | enabled, patience 5 |
| Label smoothing | 0.03 |
| Node identity embedding | 16 dimensions |
| Trainable parameters | 96,931 |

## 4. Controlled Ablation Study

### 4.1. Ablation protocol

Twenty-three controlled runs were executed. Except for the named factor, every run kept the reference settings fixed. Each ablation was capped at 12 epochs and ranked by the best validation macro-F1. This one-factor-at-a-time design makes the direction of each intervention interpretable, although it does not test all interactions among hyperparameters.

Table 3: Executed Task 3 ablations. $\Delta$ is relative to the reference validation macro-F1 of 0.6046.

| **ID** | **Changed factor** | **Val. macro-F1** | $\Delta$ |
|---|---|---|---|
| A00 | Reference: 2-layer GCN, $h = 64$, dropout 0.30 | 0.6046 | +0.0000 |
| A01 | GNN depth: 1 layer | 0.6234 | +0.0188 |
| A02 | GNN depth: 3 layers | 0.6104 | +0.0058 |
| A03 | Hidden size: 32 | 0.6016 | -0.0030 |
| A04 | Hidden size: 96 | 0.5886 | -0.0160 |
| A05 | Dropout: 0.10 | 0.6101 | +0.0055 |
| A06 | Dropout: 0.50 | 0.5845 | -0.0201 |
| A07 | Batch normalization: disabled | 0.6059 | +0.0013 |
| A08 | Learning rate: $3 \times 10^{-4}$ | 0.5877 | -0.0169 |
| **A09** | **Learning rate:** $1.5 \times 10^{-3}$ | **0.6296** | **+0.0250** |
| A10 | LR schedule: ReduceLROnPlateau | 0.6183 | +0.0137 |
| A11 | LR schedule: OneCycle | 0.6152 | +0.0106 |
| A12 | Optimizer: Adam | 0.6248 | +0.0202 |
| A13 | Optimizer: SGD, LR $5 \times 10^{-3}$ | 0.5420 | -0.0626 |
| A14 | Graph construction: anatomical only | 0.6109 | +0.0063 |
| A15 | Graph construction: functional only | 0.6090 | +0.0044 |
| A16 | Graph construction: identity graph | 0.6092 | +0.0046 |
| A17 | Edge weights: binary | 0.6106 | +0.0060 |
| A18 | GNN variant: GAT | 0.6075 | +0.0029 |
| A19 | GNN variant: GraphSAGE | 0.6210 | +0.0164 |
| A20 | Imbalance handling: weighted sampler | 0.6238 | +0.0192 |
| A21 | Imbalance handling: none | 0.6093 | +0.0047 |
| A22 | Early stopping: disabled | 0.6046 | +0.0000 |

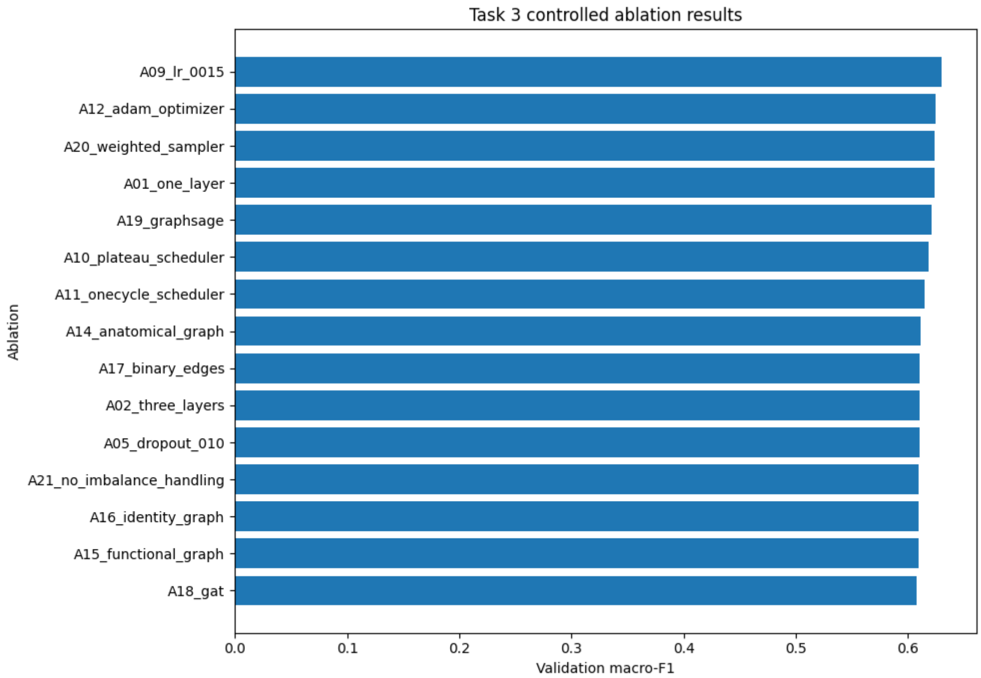



Figure 2: Top controlled ablations ranked by validation macro-F1.

### 4.2. Ablation reading

The highest-leverage factor in the executed search was optimization rather than graph topology. Raising the AdamW learning rate from $8 \times 10^{-4}$ to $1.5 \times 10^{-3}$ produced the best score, while SGD caused the largest degradation. Adam, a weighted sampler, one graph layer, and GraphSAGE also improved on the reference individually. In contrast, a 96-unit hidden layer, 0.50 dropout, and the smaller $3 \times 10^{-4}$ learning rate reduced validation macro-F1.

The graph ablations were comparatively close: anatomical-only, functional-only, identity, binary-edge, and GAT variants all remained in a narrow band around 0.607–0.611, while GraphSAGE reached 0.6210. This suggests that the main Task 2 failure was not solved by simply changing the adjacency definition. The laterality-aware readout and better optimization were more consequential. Because the ablations are single-factor tests, the notebook correctly does not claim that combining every individually positive change would necessarily improve performance.

## 5. Five-Fold Subject-Independent Results

### 5.1. Primary comparison

Table 4 gives the five-fold mean and sample standard deviation. The 1D-CNN remains the strongest model. The final Task 3 GNN improves macro-F1 by approximately 0.0940 over the first Hybrid ST-GNN, but it trails the 1D-CNN by approximately 0.1043 macro-F1.

Table 4: Five-fold subject-disjoint performance on the 91-subject development population (mean ± std).

| Model | Accuracy | Bal. Acc. | Macro-F1 | MCC | $\kappa$ | ROC-AUC | Train s | Params |
|---|---|---|---|---|---|---|---|---|
| **1D-CNN** | **0.6946 ± 0.0210** | **0.6940 ± 0.0211** | **0.6920 ± 0.0221** | **0.3940 ± 0.0409** | **0.3884 ± 0.0421** | **0.7554 ± 0.0245** | 38.50 ± 5.45 | 166,850 |
| First Hybrid ST-GNN | 0.5026 ± 0.0038 | 0.5026 ± 0.0043 | 0.4937 ± 0.0161 | 0.0052 ± 0.0090 | 0.0052 ± 0.0087 | 0.5021 ± 0.0161 | 245.69 ± 55.96 | 285,191 |
| Task 3 Laterality-Aware ST-GNN | 0.5908 ± 0.0204 | 0.5910 ± 0.0202 | 0.5877 ± 0.0196 | 0.1850 ± 0.0415 | 0.1819 ± 0.0404 | 0.6289 ± 0.0325 | 49.09 ± 13.11 | 96,931 |

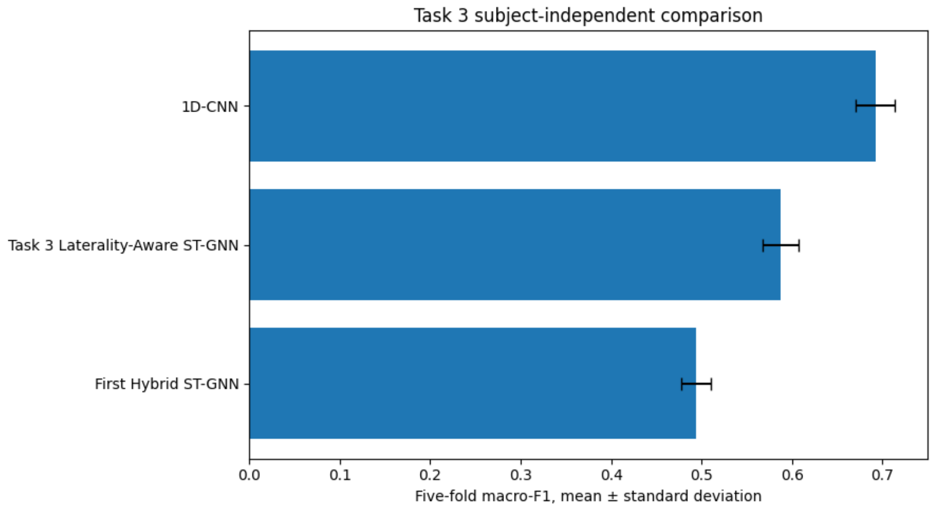



Figure 3: Five-fold macro-F1 comparison among the strongest baseline, first GNN, and final Task 3 GNN.

The fold-level macro-F1 of the final GNN was 0.6045, 0.5981, 0.5998, 0.5573, and 0.5788. The corresponding 1D-CNN values were 0.7262, 0.6996, 0.6685, 0.6853, and 0.6804. Thus the ranking is consistent across all five folds rather than being driven by one unusual partition.

### 5.2. Paired significance test versus the best baseline

Statistical inference uses subjects, not individual trials, as the unit of analysis. For each of the 91 subjects, accuracy is calculated from that subject's out-of-fold predictions for the final GNN and 1D-CNN. Let

$$d_s = \mathrm{Acc}_{s,\mathrm{GNN}} - \mathrm{Acc}_{s,\mathrm{CNN}}. \tag{2}$$

The observed mean difference is

$$\bar{d} = -0.1048. \tag{3}$$

A 10,000-repeat subject bootstrap gives a 95% confidence interval of

$$[-0.1287,\ -0.0812]. \tag{4}$$

The notebook's primary paired sign-flip randomization test uses 50,000 permutations and returns

$$p = 0.000020. \tag{5}$$

Because the confidence interval excludes zero and the permutation $p$-value is well below 0.05, the difference is statistically detectable. Its direction is unfavorable to the final GNN:

the 1D-CNN is approximately 10.5 percentage points more accurate per subject on average. Statistical significance therefore cannot be used to claim GNN superiority; it strengthens the opposite conclusion under this evaluation.

## 6. Fair Comparison with Prior Work

Published EEGMMIDB numbers vary greatly because studies use different classes, windows, subject pools, and validation protocols. Table 5 separates direct internal comparisons from literature context. Only subject-disjoint protocols should be treated as evidence about unseen-subject generalization.

Table 5: Internal and related-work context. Literature numbers are not treated as directly equivalent unless both task and validation protocol match.

| **Study/model** | **Task** | **Protocol** | **Reported result** | **Pillar-A interpretation** |
|---|---|---|---|---|
| This work: 1D-CNN | Binary left/right fist execution | 5-fold subject-disjoint CV, 91 subjects | Acc. 69.46 ± 2.10%; F1 69.20 ± 2.21% | Direct internal comparator |
| This work: final GNN | Binary left/right fist execution | 5-fold subject-disjoint CV, 91 subjects | Acc. 59.08 ± 2.04%; F1 58.77 ± 1.96% | Direct internal result |
| Bakas *et al.* (2025) [3] | Motor execution; left fist/right fist/both feet (3 classes) | 10-fold subject-independent CV; first post-cue second excluded in main analysis | Latent-aligned EEGNet balanced acc. 64.1 ± 4.3% | Subject-safe protocol, but class definition/window differ; contextual only |
| Fukushima and Miyamoto (2024) [4] | 2-class motor imagery | Cross-individual validation on PhysioNet | Accuracy 88.57% | Subject-generalization context, but imagery is a different paradigm |
| Chowdhury *et al.* (2023) [5] | Executed left/right movement | Random 70/10/20 sample split across 103 subjects | Accuracy 89.6% | Same broad execution task, but not subject-disjoint; excluded from fair ranking |

This comparison explains why apparently higher published numbers cannot automatically be used as a target for the present subject-independent experiment. For example, Chowdhury *et al.* explicitly randomly split samples across the full subject pool, so subject identity can appear in more than one partition [5]. Conversely, Bakas *et al.* used genuine subject-independent folds but solved a three-class motor-execution problem and excluded the first post-cue second because of cue-related artifacts [3]. No literature value in Table 5 is presented as an apples-to-apples replacement for the internal 1D-CNN comparison.

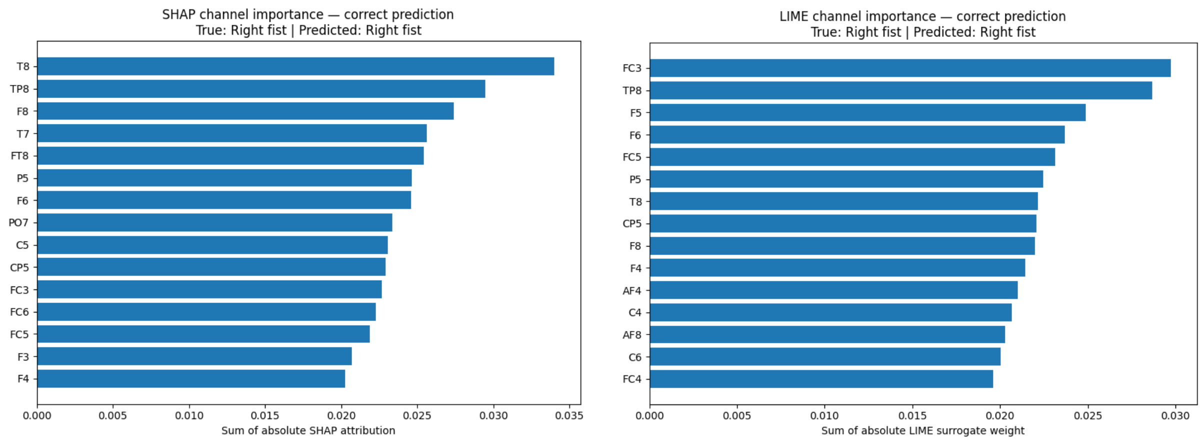
# 7. Explainability: SHAP and LIME

## 7.1. What the explanations measure

The notebook explains the interpretable $64 \times 10$ node-feature branch while keeping the raw EEG tensor of the selected trial fixed. SHAP uses `GradientExplainer` with a training-fold background. The LIME analysis is a local, distance-weighted ridge surrogate fitted to 1,000 perturbations around the selected node-feature vector. For both methods, absolute feature attributions are summed within each channel. Consequently, the figures describe *model reliance in the context of a fixed raw signal*; they do not prove neural causation or anatomical connectivity [6, 7].

## 7.2. Correct prediction

The selected correct example is from fold 4, subject 36, global trial index 1484. The true and predicted class are both right fist, with predicted right-fist probability 0.5808. SHAP ranks T8 and TP8 highest, followed by F8, T7, FT8, P5, F6, PO7, C5, and CP5 among the leading channels. The LIME surrogate places FC3 and TP8 highest, followed by F5, F6, FC5, P5, T8, CP5, and F8.




(a) SHAP, correct right-fist prediction.

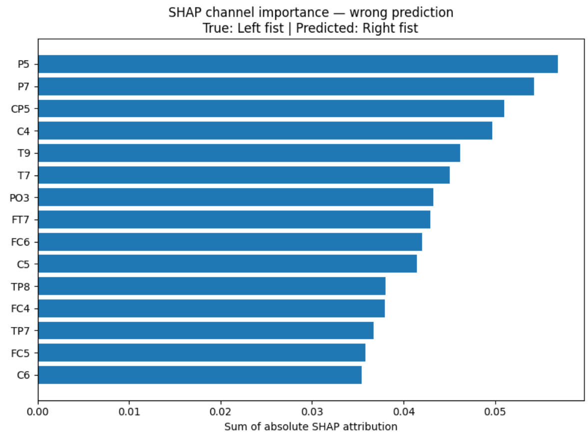
(b) LIME, correct right-fist prediction.

Figure 4: Local explanations for the correct sample.

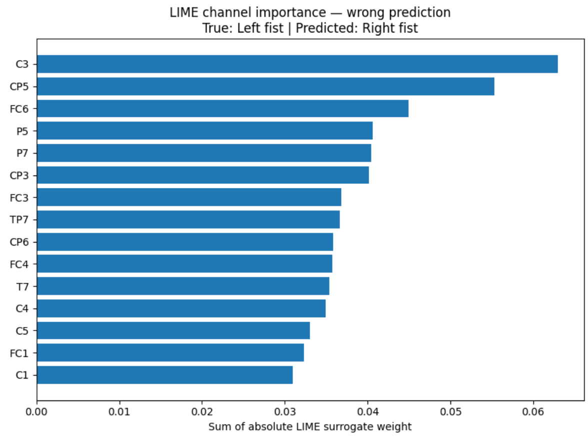
The agreement on TP8, F8, F6, P5, T8, CP5, and frontal/central channels indicates that the correct decision is supported by a distributed fronto-temporo-parietal pattern rather than a single electrode. The modest confidence (0.5808) also argues against over-interpreting the explanation as a sharply separated physiological signature.

## 7.3. Wrong prediction

The selected error is from fold 5, subject 3, global trial index 107. The true class is left fist, but the model predicts right fist with probability 0.5616. SHAP ranks P5, P7, CP5, C4, T9, T7, PO3, FT7, and FC6 among the strongest channels. LIME ranks C3, CP5, FC6, P5, P7, CP3, FC3, TP7, CP6, and FC4.




(a) SHAP, wrong left-to-right prediction.




(b) LIME, wrong left-to-right prediction.

Figure 5: Local explanations for the misclassified sample.

Here both explanation methods repeatedly emphasize central/posterior channels such as CP5, P5/P7, FC6, T7, C4/FC4 and nearby sites. Yet the predicted probability is only 0.5616. A cautious reading is that the model received an ambiguous distributed feature pattern and mapped it to the wrong hemisphere/class boundary. The plots do not justify a claim that these electrodes biologically “caused” the error.

## 8. Discussion

Task 3 succeeds at its primary engineering objective: it raises the first graph model from near-chance cross-subject performance to a reproducible macro-F1 of approximately 0.588. The new model is also substantially smaller (96,931 parameters versus 285,191) and much faster to train on average than the first Hybrid ST-GNN. The laterality-aware pooling scheme is therefore a meaningful redesign rather than a cosmetic hyperparameter change.

The ablations also provide a clear diagnostic. Learning rate, optimizer behavior, imbalance strategy, and graph depth affected performance more strongly than the precise anatomical/functional adjacency choice. This is consistent with the Task-2 observation that the initial graph model underfit. The final model learned considerably more discriminative structure, but the 1D-CNN still generalized better across subjects.

The statistically significant gap to the 1D-CNN should be treated as a result, not hidden as a failure. For this dataset and current preprocessing, the additional graph prior did not compensate for inter-subject variability as effectively as the convolutional baseline. Future work should therefore prioritize (i) fully nested subject-level tuning; (ii) subject/domain alignment; (iii) trial-adaptive or subject-adaptive graphs estimated without target labels; (iv) a controlled test of laterality pooling without graph propagation; and (v) explanation stability across many subjects rather than only two illustrative trials.

## 9. Conclusion

Task 3 completed the requested ablation, fair-comparison, five-fold CV, subject-level significance testing, and SHAP/LIME reading. Among 23 controlled ablations, increasing the AdamW learning rate to $1.5 \times 10^{-3}$ produced the best validation macro-F1 of 0.6296. The final laterality-aware ST-GNN achieved 0.5877 ± 0.0196 macro-F1 under five-fold subject-disjoint CV, improving the first Hybrid ST-GNN by about 9.4 macro-F1 points. However, the 1D-CNN remained superior at 0.6920 ± 0.0221 macro-F1. The paired subject-level randomization test confirmed that the final GNN's accuracy was significantly lower than the 1D-CNN ($p = 0.000020$, 95% bootstrap CI for the mean accuracy difference $[-0.1287, -0.0812]$).

The correct scientific conclusion is therefore: **Task 3 materially improves the first GNN, but the best leakage-conscious internal baseline remains the 1D-CNN.** The XAI examples show distributed and partially consistent channel reliance, but they should be interpreted as model explanations rather than biological causal maps. A final publication-stage experiment should make hyperparameter selection fully nested or reserve a permanent tuning set outside the outer subject folds.

## References


[1] G. Schalk, "EEG Motor Movement/Imagery Dataset," PhysioNet, Version 1.0.0, 2009, doi: 10.13026/C28G6P.

[2] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, "BCI2000: A general-purpose brain-computer interface system," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004, doi: 10.1109/TBME.2004.827072.

[3] S. Bakas, S. Ludwig, D. A. Adamos, N. Laskaris, Y. Panagakis, and S. Zafeiriou, "Latent alignment in deep learning models for EEG decoding," *Journal of Neural Engineering*, vol. 22, Art. no. 016047, 2025, doi: 10.1088/1741-2552/adb336.

[4] T. Fukushima and R. Miyamoto, "Spatiotemporal Pooling on Appropriate Topological Maps Represented as Two-Dimensional Images for EEG Classification," in *Proceedings of the Asian Conference on Computer Vision (ACCV)*, 2024, pp. 1401–1419.

[5] R. R. Chowdhury, Y. Muhammad, and U. Adeel, "Enhancing Cross-Subject Motor Imagery Classification in EEG-Based Brain–Computer Interfaces by Using Multi-Branch CNN," *Sensors*, vol. 23, no. 18, Art. no. 7908, 2023, doi: 10.3390/s23187908.

[6] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 4765–4774.

[7] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You?: Explaining the Predictions of Any Classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.